

Heat and Learning: Large-Scale Evidence on Heterogeneous Impacts Across Mexico

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Motivation

- 1 Growing evidence that heat affects academic performance
 - Park et al. (2020), Arceo-Gómez et al. (2024), Cho (2017), Garg et al. (2020)
- 2 But evidence on **heterogeneous impacts** remains limited
 - Studies with student FE → selected subpopulations
 - Studies with broad coverage → no individual-level follow-up

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- 2 But evidence on **heterogeneous impacts** remains limited
 - Studies with student FE → selected subpopulations
 - Studies with broad coverage → no individual-level follow-up
- 3 Key question: Does the effect of heat on learning vary by **climate zones**, **socioeconomic conditions**, and **student characteristics**?
- 4 Can **social protection programs** attenuate the effect?

This Paper

1 Setting: Mexico, 2007–2013 (ENLACE test)

- Census-based end-of-year exam evaluating learning over the academic year (grades 3–9)
- Nationwide panel: **18 million students**, 60+ million test scores
- Linked to daily municipal-level weather data throughout the academic year before the test

2 We organize heterogeneous effects in three dimensions: **climate zone**, **municipality characteristics**, and **student characteristics**

3 Three sets of results:

1. Heat reduces test scores, but effect varies substantially across **climate zones**
2. Effects are **heterogeneous** across municipality characteristics (poverty, rural/urban) and student characteristics (school level, baseline ability)
3. The **Progresa** conditional cash transfer can offset the negative effect of heat on learning

Results

- 1 Heat reduces test scores (-0.019 SD per 1°C increase)
 - Two channels: cumulative exposure (-0.013 SD) $>$ exam-day (-0.004 SD)
 - Warm/humid: academic-year channel; Temperate/dry: exam-day channel

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- 2 Differences in Effects Across Municipalities
 - Low SES municipalities: -0.045 SD ($8\times$ larger than high SES)
 - Rural municipalities: -0.031 SD vs. -0.007 (n.s.) in urban

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 - Elementary: cumulative exposure channel; Middle school: exam-day channel
 - Top performers lose most from cumulative heat; bottom from heat of the day of the test

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 - Elementary: cumulative exposure channel; Middle school: exam-day channel
 - Top performers lose most from cumulative heat; bottom from heat of the day of the test
- 4 Progresa (CCT): academic-year effect is -0.012 SD for non-beneficiaries, but the interaction ($+0.018$ SD) fully offsets it — net effect for beneficiaries is not significant

Outline

- 1 Data and Setting
- 2 Empirical Strategy
- 3 Main Results: Temperature Effects by Climate Zone
 - Heterogeneous Effects: Municipalities
 - Heterogeneous Effects: Students
- 4 Conditional Cash Transfer & Heat
- 5 Conclusions

Data: ENLACE + Weather

- 1 ENLACE: national standardized test (math and reading), grades 3–9, 2007–2013
 - Census-based end-of-year exam evaluating learning over the academic year (grades 3–9)
 - Follow same students across years.
 - >18 million unique students; >60 million student×year observations

- 2 Daily weather data interpolated to municipality level (2,258 municipalities).
 - Key variable: average daily max temperature during academic year
 - Also: exam-day temperature, precipitation

- 3 Progresá: individual-level indicator of CCT receipt (grades 6–9)
 - 26.8% of students are beneficiaries

Data: Student Panel Structure

Academic Year	3	4	5	6	7	8	9
2007–2008	C1	C2	C3	C4			
2008–2009	C5	C1	C2	C3	C4		
2009–2010	C6	C5	C1	C2	C3	C4	
2010–2011		C6	C5	C1	C2	C3	C4
2011–2012			C6	C5	C1	C2	C3
2012–2013				C6	C5	C1	C2

- Cohorts C1–C6 are tracked longitudinally across grades.
- Identification exploits within-student variation in temperature exposure across academic years.

Data: Geographic Variation in Temperature

Number of days above 28°C during the **2008** academic year.



Days > 28°C

0

1-60

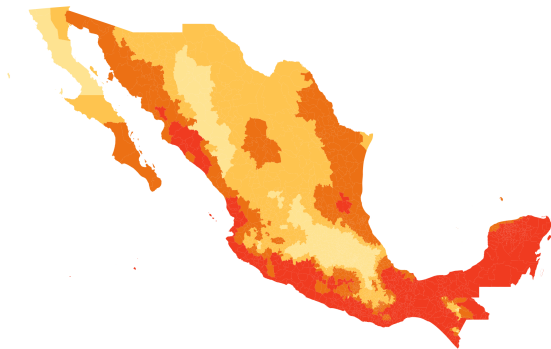
61-120

121-180

181-259

Data: Geographic Variation in Temperature

Number of days above 28°C during the **2010** academic year.



Days > 28°C

0

1-60

61-120

121-180

181-259

Data: Geographic Variation in Temperature

Number of days above 28°C during the **2012** academic year.



Days > 28°C

0

1-60

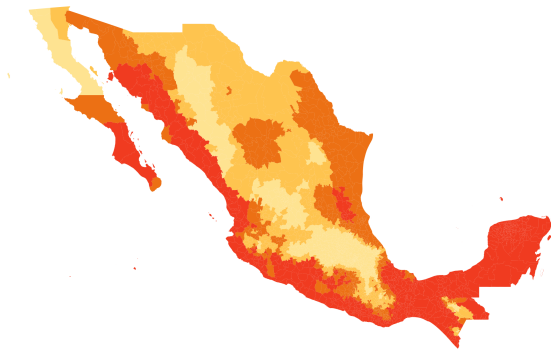
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Data: Geographic Variation in Temperature

Number of days above 28°C during the **2014** academic year.



Days > 28°C

0

1-60

61-120

121-180

181-259

Data: Municipalities by Climate Zone

Descriptives

- The two classifications overlap: $\sim 40\%$ of dry municipalities are temperate, and $\sim 60\%$ are warm.

	All	Warm	Temperate	Dry	Humid
Mean max temp, full year ($^{\circ}\text{C}$)	26.9	29.0	24.4	26.2	27.2
Days $>34^{\circ}\text{C}$ per school year	15.8	28.7	1.5	17.3	15.0



Moisture Classification Dry Humid

(a) Dry/Humid



Temperature Classification Temperate Warm

(b) Warm/Temperate

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Empirical Strategy

- Student fixed effects regression exploiting within-student variation in temperature:

$$test_{ismgy} = \beta T_{my} + \delta P_{my} + \eta_i + \rho_m + \gamma_y + \phi_g + \tau_{sy} + \epsilon_{ismgy}$$

T_{my} : avg daily max temp during academic year

η_i : **student fixed effects** — within-student variation across years

ρ_m : municipality FE; γ_y : year FE; ϕ_g : grade FE

τ_{sy} : state $\times$ linear time trends

Standard errors clustered at municipality level

Empirical Strategy

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Standard errors clustered at municipality level

- Extended: decompose into academic-year + exam-day temperature
- Estimated **separately by climate type** (warm/temperate, dry/humid)

Empirical Strategy

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Standard errors clustered at municipality level

- Extended: decompose into academic-year + exam-day temperature
- Estimated **separately by climate type** (warm/temperate, dry/humid)
- **Interpretation:** $\hat{\beta}$ = SD change in test scores for a 1°C increase in avg. daily max temperature during the academic year

Empirical Strategy: Binned Specification

- Replace continuous temperature with days in different temperature ranges:

$$\begin{aligned} test_{ismgy} = & \beta_1 Days_{>33} + \beta_2 Days_{30-33} + \beta_3 Days_{26-30} + \beta_4 Days_{22-26} \\ & + \beta_5 Days_{<18} + \delta P_{my} + \eta_i + \rho_m + \gamma_y + \phi_g + \tau_{sy} + \epsilon_{ismgy} \end{aligned}$$

- Omitted category: 18–22°C (thermal comfort zone)
- Each β_k : effect of one additional day in bin k vs. one day at 18–22°C
- Estimated **separately by climate type** to avoid pooling bias (Jones et al. 2026)

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Main Results: Temperature and Test Scores FE Lag Heat Index

- Higher temperatures over the full academic year lower test scores (-0.019 SD/ $^{\circ}\text{C}$), especially in warm and humid climates.

	(1) All Sample	(2) Warm	(3) Temperate	(4) Dry	(5) Humid
<i>Panel A: Avg Max Temp ($^{\circ}\text{C}$)</i>					
Full Academic Year	-0.019^{***} (0.004)				
<i>Panel B: Exam Day Temp ($^{\circ}\text{C}$)</i>					
Full Academic Year	-0.013^{***} (0.004)				
Exam Day	-0.004^{***} (0.001)				
Observations	64.0M				
Mean Temp ($^{\circ}\text{C}$)	26.9				
SD of Temp ($^{\circ}\text{C}$)	2.96				
Within-FE SD ($^{\circ}\text{C}$)	0.34				

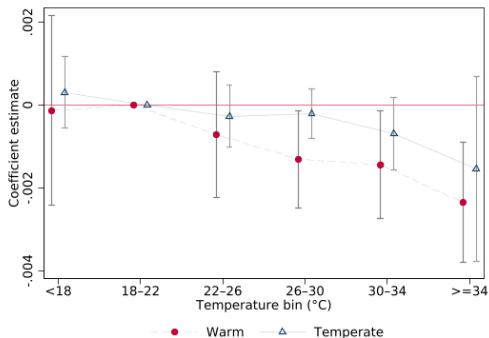
Main Results: Temperature and Test Scores FE Lag Heat Index

- Higher temperatures over the full academic year lower test scores (-0.019 SD/ $^{\circ}\text{C}$), especially in warm and humid climates.
- In temperate and dry climates, exam-day temperature is the more relevant margin.

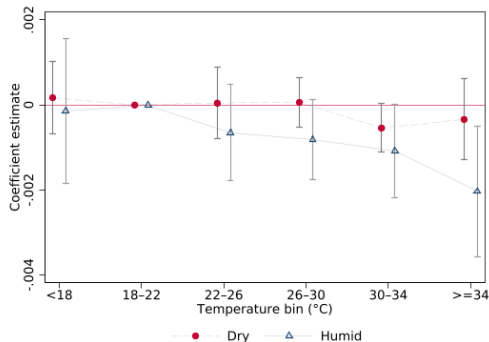
	(1) All Sample	(2) Warm	(3) Temperate	(4) Dry	(5) Humid
<i>Panel A: Avg Max Temp ($^{\circ}\text{C}$)</i>					
Full Academic Year	-0.019^{***} (0.004)	-0.023^{***} (0.005)	-0.012^{**} (0.005)	-0.010^{**} (0.004)	-0.021^{***} (0.005)
<i>Panel B: Exam Day Temp ($^{\circ}\text{C}$)</i>					
Full Academic Year	-0.013^{***} (0.004)	-0.023^{***} (0.006)	-0.002 (0.005)	-0.003 (0.004)	-0.020^{***} (0.007)
Exam Day	-0.004^{***} (0.001)	-0.000 (0.002)	-0.006^{***} (0.002)	-0.006^{***} (0.002)	-0.001 (0.002)
Observations	64.0M	33.6M	30.3M	22.5M	41.2M
Mean Temp ($^{\circ}\text{C}$)	26.9	29.1	24.4	26.2	27.3
SD of Temp ($^{\circ}\text{C}$)	2.96	1.40	2.15	2.60	3.07
Within-FE SD ($^{\circ}\text{C}$)	0.34	0.37	0.28	0.37	0.30

Nonlinear Effects: Warm vs. Temperate

- The effect is nonlinear in warm and humid climates, with large effects concentrated at extreme temperatures; in temperate and dry climates the decline is more gradual.



(a) Warm and Temperate Weather



(b) Dry and Humid Weather

Coefficients from binned specification. Omitted category: 18–22°C. 95% CI shown.

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Heterogeneity: Socioeconomic Status (marginalization index)

- Effect is **8× larger** in low SES vs. high SES municipalities.

	All	Warm	Temperate	Dry	Humid
<i>High SES</i>					
Acad. Year Temp	−0.006 (0.005)				
Exam Day Temp	−0.006*** (0.001)				
Observations	45.5M				
<i>Medium SES</i>					
Acad. Year Temp	−0.016** (0.007)				
Exam Day Temp	0.000 (0.002)				
Observations	11.9M				
<i>Low SES</i>					
Acad. Year Temp	−0.045*** (0.013)				
Exam Day Temp	0.005 (0.005)				
Observations	6.3M				

Heterogeneity: Socioeconomic Status (marginalization index)

- Effect is **8× larger** in low SES vs. high SES municipalities.

	All	Warm	Temperate	Dry	Humid
<i>High SES</i>					
Acad. Year Temp	−0.006 (0.005)	−0.016** (0.007)	0.006 (0.007)	−0.004 (0.005)	−0.008 (0.010)
Exam Day Temp	−0.006*** (0.001)	−0.001 (0.002)	−0.008*** (0.002)	−0.007*** (0.002)	−0.005* (0.003)
Observations	45.5M	21.7M	23.7M	18.8M	26.5M
<i>Medium SES</i>					
Acad. Year Temp	−0.016** (0.007)	−0.027*** (0.009)	0.005 (0.009)	−0.005 (0.009)	−0.014 (0.009)
Exam Day Temp	0.000 (0.002)	0.003 (0.003)	−0.007** (0.003)	0.001 (0.003)	−0.000 (0.003)
Observations	11.9M	7.0M	4.9M	3.4M	8.5M
<i>Low SES</i>					
Acad. Year Temp	−0.045*** (0.013)	−0.061*** (0.017)	−0.012 (0.018)	0.032 (0.021)	−0.046*** (0.014)
Exam Day Temp	0.005 (0.005)	0.009 (0.006)	0.001 (0.004)	−0.003 (0.007)	0.005 (0.005)
Observations	6.3M	4.7M	1.7M	0.3M	6.0M

Heterogeneity: Rural vs. Urban

- Rural: cumulative exposure channel. Urban: primarily exam-day channel.
- Suggests urban infrastructure buffers cumulative but not exam-day exposure.

	All	Warm	Temperate	Dry	Humid
<i>Rural</i>					
Acad. Year Temp	-0.031*** (0.008)				
Exam Day Temp	0.004 (0.003)				
Observations	12.0M				
<i>Urban</i>					
Acad. Year Temp	-0.007 (0.004)				
Exam Day Temp	-0.006*** (0.001)				
Observations	51.8M				

Heterogeneity: Rural vs. Urban

- Rural: cumulative exposure channel. Urban: primarily exam-day channel.
- Suggests urban infrastructure buffers cumulative but not exam-day exposure.

	All	Warm	Temperate	Dry	Humid
<i>Rural</i>					
Acad. Year Temp	-0.031*** (0.008)	-0.039*** (0.011)	-0.006 (0.010)	-0.003 (0.009)	-0.028*** (0.010)
Exam Day Temp	0.004 (0.003)	0.005 (0.003)	-0.002 (0.003)	0.001 (0.003)	0.003 (0.003)
Observations	12.0M	7.6M	4.4M	2.6M	9.3M
<i>Urban</i>					
Acad. Year Temp	-0.007 (0.004)	-0.016*** (0.006)	0.002 (0.005)	-0.004 (0.004)	-0.008 (0.008)
Exam Day Temp	-0.006*** (0.001)	-0.003 (0.002)	-0.007*** (0.002)	-0.006*** (0.002)	-0.006** (0.002)
Observations	51.8M	25.9M	25.8M	19.8M	31.8M

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Heterogeneity: Elementary vs. Middle School

- Elementary: cumulative exposure matters. Middle school: exam-day effect dominates.
- Positive acad-year coefficients for middle school may reflect selective attrition.

	All	Warm	Temperate	Dry	Humid
<i>Elementary School (grades 3–6)</i>					
Acad. Year Temp	−0.012** (0.005)				
Exam Day Temp	−0.004** (0.002)				
Observations	38.5M				
<i>Middle School (grades 7–9)</i>					
Acad. Year Temp	0.015 (0.008)				
Exam Day Temp	−0.010*** (0.003)				
Observations	20.8M				

Heterogeneity: Elementary vs. Middle School

- Elementary: cumulative exposure matters. Middle school: exam-day effect dominates.
- Positive acad-year coefficients for middle school may reflect selective attrition.

	All	Warm	Temperate	Dry	Humid
<i>Elementary School (grades 3–6)</i>					
Acad. Year Temp	−0.012** (0.005)	−0.021*** (0.006)	0.005 (0.005)	0.001 (0.005)	−0.024*** (0.006)
Exam Day Temp	−0.004** (0.002)	−0.001 (0.003)	−0.007*** (0.001)	−0.005*** (0.002)	0.000 (0.002)
Observations	38.5M	20.0M	18.4M	13.7M	24.6M
<i>Middle School (grades 7–9)</i>					
Acad. Year Temp	0.015 (0.008)	−0.011 (0.010)	0.025** (0.012)	0.008 (0.008)	0.027** (0.012)
Exam Day Temp	−0.010*** (0.003)	−0.002 (0.004)	−0.013*** (0.005)	−0.013*** (0.004)	−0.013*** (0.003)
Observations	20.8M	11.0M	9.7M	7.2M	13.5M

Heterogeneity: Baseline Academic Performance

- Cumulative effect largest for **top tercile** (more to lose); exam-day effect largest for **bottom tercile**.

	All	Warm	Temperate	Dry	Humid
<i>Bottom tercile</i>					
Acad. Year	−0.011** (0.005)				
Exam Day	−0.006*** (0.002)				
Observations	20.4M				
<i>Middle tercile</i>					
Acad. Year	−0.015*** (0.005)				
Exam Day	−0.003* (0.002)				
Observations	21.5M				
<i>Top tercile</i>					
Acad. Year	−0.015*** (0.005)				
Exam Day	−0.002 (0.001)				
Observations	22.1M				

Heterogeneity: Baseline Academic Performance

- Cumulative effect largest for **top tercile** (more to lose); exam-day effect largest for **bottom tercile**.

	All	Warm	Temperate	Dry	Humid
<i>Bottom tercile</i>					
Acad. Year	-0.011** (0.005)	-0.013* (0.007)	-0.007 (0.006)	0.002 (0.008)	-0.018** (0.008)
Exam Day	-0.006*** (0.002)	-0.003 (0.002)	-0.008*** (0.002)	-0.009*** (0.003)	-0.004 (0.003)
Observations	20.4M	10.7M	9.6M	7.2M	13.1M
<i>Middle tercile</i>					
Acad. Year	-0.015*** (0.005)	-0.025*** (0.007)	-0.002 (0.005)	-0.003 (0.004)	-0.021** (0.008)
Exam Day	-0.003* (0.002)	0.001 (0.002)	-0.006*** (0.002)	-0.006*** (0.002)	-0.000 (0.003)
Observations	21.5M	11.3M	10.2M	7.6M	13.9M
<i>Top tercile</i>					
Acad. Year	-0.015*** (0.005)	-0.031*** (0.006)	0.002 (0.005)	-0.010** (0.005)	-0.020*** (0.007)
Exam Day	-0.002 (0.001)	0.001 (0.002)	-0.004*** (0.001)	-0.003** (0.002)	0.002 (0.002)
Observations	22.1M	11.6M	10.4M	7.8M	14.3M

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Progresa: Can Social Protection Buffer Heat?

- Progresa (Oportunidades): Mexico's flagship CCT
 - Income support conditional on school attendance
 - Individual-level receipt indicator linked to test data (grades 6–9)
 - Minimal selection concerns given student fixed effects
- Specification:

$$test_{ismgy} = \beta T_{my} + \theta CCT_{iy} + \lambda CCT_{iy} \times T_{my} + \beta_2 T_{my}^{exam} + FE + \epsilon$$

- λ : differential temperature effect for Progresa recipients
- $\beta + \lambda$: total temperature effect for beneficiaries

Progresa: Results

- Progresa **nearly fully offsets** the cumulative heat effect. Marginal effect for beneficiaries is indistinguishable from zero.
- Exam-day effect unaffected — consistent with CCT operating through sustained channels.

	All	Warm	Temperate	Dry	Humid
Avg Max Temp (°C)	-0.012** (0.005)	-0.029*** (0.008)	-0.004 (0.006)	-0.009* (0.005)	-0.011 (0.010)
Progresa × Avg Max Temp	0.018*** (0.002)	0.032*** (0.004)	0.025*** (0.004)	0.013*** (0.003)	0.020*** (0.002)
Exam Day Temp	-0.006*** (0.002)	-0.001 (0.002)	-0.008** (0.003)	-0.007*** (0.002)	-0.007** (0.003)
<i>Net effect: beneficiaries</i>	0.006 (0.005)	0.003 (0.007)	0.021*** (0.007)	0.004 (0.006)	0.009 (0.010)
Observations	31.4M	16.6M	14.6M	10.9M	20.3M

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Conclusions

- 1 Heat reduces learning, but the average effect **masks substantial heterogeneity**
 - Largest effects in warm, humid, poor, rural municipalities
 - Elementary students: cumulative exposure; Middle school: exam-day
 - Top performers lose most from cumulative heat; bottom from heat in the exam period
- 2 Social protection as climate adaptation
 - Progresa nearly fully offsets the cumulative effect for beneficiaries
 - Scalable complement to infrastructure-based interventions
- 3 Policy implication: uniform responses to heat will miss the populations most at risk — **context matters**

Thank you!

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Data: Summary Statistics [Back](#)

	All	Warm	Temperate	Dry	Humid
<i>Panel A: Education & Climate</i>					
Avg standardized score	0.008	0.007	0.010	0.014	0.005
Share scholarship recipients	0.268	0.320	0.211	0.191	0.310
Mean max temp, full year (°C)	26.9	29.0	24.4	26.2	27.2
Days >34°C per school year	15.8	28.7	1.5	17.3	15.0
<i>Panel B: Municipality Characteristics</i>					
Rural population (%)	23.4	27.8	18.5	18.2	26.2
Illiteracy rate, 15+ (%)	7.0	8.3	5.4	4.7	8.2
No piped water (%)	11.4	14.4	8.0	6.4	14.0
Air conditioning in schools	22.9	35.2	10.0	31.8	17.9
N municipalities	2,268	1,406	862	613	1,655
Students per year (approx.)	10.7M	5.7M	5.1M	3.8M	7.0M

Main Results: Adding FE [Back](#)

- Academic-year effect emerges after municipality FE, highlighting the importance of within-municipality variation.
- Estimates are stable across specifications (cols 2–4); exam-day effect attenuates with state $\times$ year trends.

	(1)	(2)	(3)	(4)
Academic Year	0.0032 (0.0030)	-0.0145*** (0.0047)	-0.0146*** (0.0047)	-0.0133*** (0.0044)
Exam Day	-0.0138*** (0.0020)	-0.0130*** (0.0018)	-0.0124*** (0.0018)	-0.0036*** (0.0013)
Student FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Municipality FE		Yes	Yes	Yes
Grade FE			Yes	Yes
State $\times$ Year				Yes
Observations	64,005,332	64,005,332	64,005,332	64,005,332

Robustness: The Effect Is Contemporaneous [Back](#)

- Future temperatures do not predict current scores, ruling out confounding trends.

	All	Warm	Temperate	Dry	Humid
Acad-Year Temp ($t - 1$)	-0.001 (0.004)	0.006 (0.005)	-0.011 (0.008)	0.004 (0.005)	-0.008 (0.008)
Acad-Year Temp (t)	-0.011** (0.004)	-0.024*** (0.008)	-0.005 (0.005)	-0.005 (0.005)	-0.027** (0.010)
Exam-Day Temp	-0.004*** (0.001)	0.000 (0.002)	-0.006*** (0.002)	-0.006*** (0.002)	-0.001 (0.002)
Acad-Year Temp ($t + 1$)	-0.005 (0.005)	-0.007 (0.006)	-0.003 (0.006)	-0.003 (0.004)	-0.012 (0.008)

Robustness: Heat Index [Back](#)

- The Heat Index combines temperature and relative humidity into a measure of perceived heat stress.
- Results are consistent with the main specification, with slightly larger estimates reflecting the added humidity channel.

	(1) All Sample	(2) Warm	(3) Temperate	(4) Dry	(5) Humid
Panel A: Avg HI (°C)					
Full Academic Year	-0.0258*** (0.0046)	-0.0242*** (0.0064)	-0.0229*** (0.0056)	-0.0141*** (0.0053)	-0.0343*** (0.0070)
Panel B: Exam Day HI (°C)					
Full Academic Year	-0.0242*** (0.0047)	-0.0240*** (0.0063)	-0.0141*** (0.0053)	-0.0103* (0.0055)	-0.0335*** (0.0073)
Exam day	-0.0020 (0.0013)	-0.0010 (0.0014)	-0.0060*** (0.0019)	-0.0062*** (0.0015)	-0.0009 (0.0016)
Observations	64,005,332	33,576,955	30,251,695	22,513,653	41,240,101
Mean HI	17.2	20.7	13.3	16.2	17.7
SD	0.30	0.33	0.25	0.32	0.26